

Answers from Q100 Onwards

Found 19 answers.

ID: Q100

Question: What is the primary treatment for Stage IVC laryngeal cancer?

Expected Answer: The intent is palliative treatment for symptom relief.

Generated Answer: Patients with Stage IVc laryngeal cancer are candidates for palliative treatment, with decisions regarding local radiotherapy or systemic treatment based on individual clinical situations.

ID: Q101

Question: A 76-year-old African American man presents to his primary care provider complaining of urinary frequency. He wakes up 3-4 times per night to urinate while he previously only had to wake up once per night. He also complains of post-void dribbling and difficulty initiating a stream of urine. He denies any difficulty maintaining an erection. His past medical history is notable for non-alcoholic fatty liver disease, hypertension, hyperlipidemia, and gout. He takes aspirin, atorvastatin, enalapril, and allopurinol. His family history is notable for prostate cancer in his father and lung cancer in his mother. He has a 15-pack-year smoking history and drinks alcohol socially. On digital rectal exam, his prostate is enlarged, smooth, and non-tender. Which of the following medications is indicated in this patient?

Expected Answer: Tamsulosin

Generated Answer: Enalapril is indicated in this patient for the management of benign prostatic hyperplasia (BPH), particularly given his hypertension and urinary symptoms. However, it is essential to continue monitoring his PSA levels and consider biopsy if indicated, given his age, symptoms, and family history of prostate cancer.

ID: Q102

Question: A previously healthy 36-year-old man comes to the physician for a yellow discoloration of his skin and dark-colored urine for 2 weeks. He does not drink any alcohol. Physical examination shows jaundice. Abdominal and neurologic examinations show no abnormalities. Serum studies show increased levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST). A liver biopsy is performed and a photomicrograph after periodic acid-Schiff-staining is shown. Which of the following is the most likely additional finding in this patient?

Expected Answer: Bullous changes of the lung bases on chest CT

Generated Answer: Based on the provided clinical context, specifically Source 2 and Source 4, there is no direct evidence to correlate the patient's symptoms and liver biopsy findings with hepatoblastoma or small bowel malignancy. The patient's presentation with jaundice,

dark-colored urine, and elevated ALT and AST levels suggests a primary liver or biliary tract pathology. Given the periodic acid-Schiff-staining photomicrograph, which likely shows characteristic glycogen accumulation, the most likely diagnosis is hepatocellular carcinoma (HCC). HCC in adults is histologically similar to that seen in children, as mentioned in Source 2. In Source 4, it is noted that gastrointestinal symptoms develop in 75% of patients with malignant lesions of the small bowel, but the patient's presentation does not fit this profile. Thus, the most likely additional finding in this patient would be common bile duct obstruction from an HCC, which could present with jaundice and abdominal pain.

ID: Q103

Question: A 69-year-old male presents to the emergency room with back pain. He has a history of personality disorder and metastatic prostate cancer and was not a candidate for surgical resection. He began chemotherapy but discontinued due to unremitting nausea. He denies any bowel or bladder incontinence. He has never had pain like this before and is demanding morphine. The nurse administers IV morphine and he feels more comfortable. Vital signs are stable. On physical examination you note tenderness to palpation along the lower spine, weakness in the bilateral lower extremities, left greater than right. Neurological examination is also notable for hyporeflexia in the knee and ankle jerks bilaterally. You conduct a rectal examination, which reveals saddle anesthesia. Regarding this patient, what is the most likely diagnosis and the appropriate next step in management?

Expected Answer: The most likely diagnosis is cauda equina syndrome and steroids should be started prior to MRI

Generated Answer: The most likely diagnosis for this 69-year-old male patient presenting with back pain, tenderness along the lower spine, bilateral lower extremity weakness with left greater than right, hyporeflexia in knee and ankle jerks, and saddle anesthesia upon rectal examination is metastatic spinal cord compression due to his history of prostate cancer. Given the lack of bowel or bladder incontinence, it suggests an epidural metastasis causing compression of the spinal cord, rather than a cauda equina syndrome.

The appropriate next step in management is to urgently order a magnetic resonance imaging (MRI) of the spine to confirm the diagnosis and localize the extent of compression. While waiting for imaging results, continue administering IV morphine for pain control and consider starting dexamethasone at 10-100 mg intravenously, followed by 4-24 mg every 4-6 hours to stabilize or improve neurological function until definitive treatment. Consultation with a multidisciplinary team including oncology, radiation oncology, neurosurgery, and palliative care is recommended to determine the optimal treatment plan, which may include radiation therapy, surgery, or continued chemotherapy, depending on the patient's overall health status and cancer progression.

ID: Q104

Question: A 52-year-old woman comes to the physician because of a 6-month history of generalized fatigue, low-grade fever, and a 10-kg (22-lb) weight loss. Physical examination shows generalized pallor and splenomegaly. Her hemoglobin concentration is 7.5 g/dL and leukocyte count is 41,800/mm³. Leukocyte alkaline phosphatase activity is low. Peripheral blood smear shows basophilia with myelocytes and metamyelocytes. Bone marrow biopsy shows cellular hyperplasia with proliferation of immature granulocytic cells. Which of the following mechanisms is most likely responsible for this patient's condition?

Expected Answer: Unregulated expression of the ABL1 gene

Generated Answer: The patient's condition is most likely chronic myelogenous leukemia (CML), characterized by a clonal proliferation of myelogenous stem cells, often presenting with fatigue, fever, weight loss, pallor, and splenomegaly. The low leukocyte alkaline phosphatase activity and peripheral blood smear findings of basophilia with myelocytes and metamyelocytes support this diagnosis. The bone marrow biopsy showing cellular hyperplasia with proliferation of immature granulocytic cells further confirms the diagnosis. The most likely mechanism responsible for this patient's condition is the presence of the Philadelphia chromosome, resulting from a reciprocal translocation between chromosomes 9 and 22, which creates a fusion gene (BCR-ABL1) with constitutively active tyrosine kinase activity. This genetic abnormality drives the uncontrolled proliferation of myeloid cells in CML.

ID: Q105

Question: A 5-year-old girl is brought to the clinic by her mother for excessive hair growth. Her mother reports that for the past 2 months she has noticed hair at the axillary and pubic areas. She denies any family history of precocious puberty and reports that her daughter has been relatively healthy with an uncomplicated birth history. She denies any recent illnesses, weight change, fever, vaginal bleeding, pain, or medication use. Physical examination demonstrates Tanner stage 4 development. A pelvic ultrasound shows an ovarian mass. Laboratory studies demonstrate an elevated level of estrogen. What is the most likely diagnosis?

Expected Answer: Granulosa cell tumor

Generated Answer: The most likely diagnosis for a 5-year-old girl presenting with excessive hair growth (Tanner stage 4 development), accompanied by an elevated estrogen level and an ovarian mass identified on pelvic ultrasound, is a sex cord stromal tumor (SCST) of the ovary, specifically a juvenile granulosa cell tumor (GCT). These tumors are rare but can occur in children and adolescents, and their management often involves surgical removal of the tumor with appropriate staging for malignancy, given the potential for 10%-20% of ovarian tumors in children to be malignant.

ID: Q106

Question: A 70-year-old man presents to a medical clinic reporting blood in his urine and lower abdominal pain for the past few days. He is also concerned about urinary frequency and urgency. He states that he recently completed a cycle of chemotherapy for non-Hodgkin lymphoma. Which medication in the chemotherapy regimen most likely caused his symptoms?

Expected Answer: Cyclophosphamide

Generated Answer: The medication in the chemotherapy regimen most likely causing the patient's symptoms is cyclophosphamide, as it is an alkylating agent known to cause hemorrhagic cystitis, which includes symptoms such as hematuria, dysuria, and bladder pain. Cyclophosphamide's toxic metabolite, acrolein, is believed to be responsible for urothelial toxicity. This patient's recent chemotherapy cycle for non-Hodgkin lymphoma, along with his presenting symptoms, strongly suggest cyclophosphamide as the likely culprit.

ID: Q107

Question: A 38-year-old woman undergoes hemithyroidectomy for treatment of localized, well-differentiated papillary thyroid carcinoma. The lesion is removed with clear margins. However, during the surgery, a structure lying directly adjacent to the superior thyroid artery at the upper pole of the thyroid lobe is damaged. This patient is most likely to experience which of the following symptoms?

Expected Answer: Voice pitch limitation

Generated Answer: The damaged structure most likely to cause symptoms in this patient is the external branch of the superior laryngeal nerve. This nerve has a variable course and is intimately associated with the superior thyroid vessels in approximately 58% of cases, either by traveling between branches of the superior thyroid artery, running in close proximity (< 1 cm) superior to the vessels, or coursing posterior to the vessels and superior pole of the thyroid gland. Injury to this nerve can result in difficulty with high-pitched tones due to its function in controlling the cricothyroid muscle. This nerve damage would not typically cause significant bleeding, but could lead to voice changes and vocal cord dysfunction.

ID: Q108

Question: A 55-year-old man comes to the physician because of fatigue and worsening abdominal pain for 4 weeks. He also reports excessive night sweats and a 5.4-kg (12-lb) weight loss during this time. He has a neck swelling for 4 days. Physical examination shows a nontender, enlarged, and fixed supraclavicular lymph node. There is splenomegaly. A CT scan of the thorax and abdomen shows massively enlarged axillary, mediastinal, and cervical lymph nodes. Analysis of an excised cervical lymph node shows lymphocytes with a high proliferative index that stain positive for CD20. Which of the following is the most likely diagnosis?

Expected Answer: Diffuse large B-cell lymphoma

Generated Answer: The most likely diagnosis for this 55-year-old man presenting with fatigue, abdominal pain, weight loss, night sweats, and a neck swelling along with findings of a nontender, enlarged, and fixed supraclavicular lymph node, splenomegaly, and massively enlarged axillary, mediastinal, and cervical lymph nodes on CT scan, and lymphocytes with a high proliferative index staining positive for CD20, is nodular lymphocyte-predominant Hodgkin lymphoma (NLPHL). This conclusion is supported by the German Hodgkin Lymphoma Study Group's large retrospective study of 394 patients with NLPHL [Source 1], which reported that patients were predominantly male (75%), only 9% had B symptoms, and 79% had early-stage disease, with tumor control (freedom from treatment failure) and overall survival being 88% and 96%, respectively, slightly better than in classical Hodgkin lymphoma (cHL) (82% and 92%, respectively) [Source 1]. Additionally, the immunophenotypic profile of CD20 positivity in the excised cervical lymph node supports the diagnosis of B-cell NHL, which NLPHL belongs to.

ID: Q109

Question: A 57-year-old man comes to the physician for a follow-up evaluation of chronic, retrosternal chest pain. The pain is worse at night and after heavy meals. He has taken oral

pantoprazole for several months without any relief of his symptoms. Esophagogastroduodenoscopy shows ulcerations in the distal esophagus and a proximally dislocated Z-line. A biopsy of the distal esophagus shows columnar epithelium with goblet cells. Which of the following microscopic findings underlie the same pathomechanism as the cellular changes seen in this patient?

Expected Answer: Squamous epithelium in the bladder

Generated Answer: The microscopic findings that underlie the same pathomechanism as the cellular changes seen in this patient, characterized by columnar epithelium with goblet cells in the distal esophagus, are intestinal metaplasia and dysplasia. These changes occur as a result of chronic gastroesophageal reflux disease (GERD) leading to the replacement of normal stratified squamous epithelium in the esophagus with metaplastic columnar epithelium resembling that of the small intestine, including goblet cells. This process is associated with an increased risk of adenocarcinoma development, as mentioned in Source 1 and Source 2.

ID: Q110

Question: A 53-year-old woman with hypertension and hyperlipidemia comes to the physician because of generalized reddening of her skin and itching for the past 2 weeks. Her symptoms occur every evening before bedtime and last for about 30 minutes. Three months ago, atorvastatin was stopped after she experienced progressively worsening neck and back pain. Statin therapy was reinitiated at lower doses 3 weeks ago but had to be stopped again after her musculoskeletal symptoms recurred. Her menses occur irregularly at 2–3 month intervals and last for 3–4 days. She has smoked one pack of cigarettes daily for the past 30 years. Her current medications include lisinopril and niacin. Her brother died of colonic adenocarcinoma, and her father died of small cell lung cancer. She is 169 cm (5 ft 6 in) tall and weighs 83 kg (183 lb); BMI is 29 kg/m². Her vital signs are within normal limits. Physical examination shows no abnormalities. Serum lipid studies show: Total cholesterol 247 mg/dL HDL-cholesterol 39 mg/dL LDL-cholesterol 172 mg/dL Triglycerides 152 mg/dL Which of the following is the most appropriate next step in management?"

Expected Answer: Administer ibuprofen

Generated Answer: The most appropriate next step in management for this patient would be to consider dermatologic evaluation and possibly a skin biopsy to investigate the cause of her generalized reddening and itching, particularly given her recent statin exposure and history of musculoskeletal symptoms. This could potentially be related to drug-induced lupus erythematosus, erythema gyratum repens (as mentioned in Source 1), or another condition. Additionally, considering her family history and smoking, it is important to continue screening for potential underlying malignancies, such as breast cancer or lymphoma, which may also present with dermatologic symptoms.

ID: Q111

Question: A 45-year-old man is transferred to the intensive care unit from the emergency department for acute respiratory failure. He was rushed to the hospital after developing progressive respiratory distress over the last 24 hours. His medical history is significant for long-standing severe persistent asthma, hypertension, and several bouts of community and hospital-acquired pneumonia. His medications include amlodipine, lisinopril, inhaled

fluticasone, salmeterol, and oral prednisone. He is a lifelong non-smoker and drinks alcohol occasionally on the weekends. He works as a sales executive and went to Hawaii a month ago. In the emergency department, he was started on broad-spectrum antibiotics and bronchodilators. His respiratory failure progressively worsens, and on day 2 of admission, he requires mechanical ventilator support. Chest X-ray shows multiple nodules bilaterally in the lower lobes. Flexible bronchoscopy is performed and the bronchoalveolar lavage sample from the medial segment of the right lower lobe shows neutrophils, and the fungal preparation shows *Aspergillus fumigatus*. A video-assisted thoracoscopy is performed and biopsy from the right lower lobe is taken which shows plugging of the terminal bronchioles with mucus, inflammatory cells, and fungal vascular invasion. Which of the following is the most likely mechanism responsible for the biopsy findings?

Expected Answer: Defects in the immune response

Generated Answer: The most likely mechanism responsible for the biopsy findings in this 45-year-old man with acute respiratory failure, severe persistent asthma, and community and hospital-acquired pneumonia, presenting with multiple nodules on chest X-ray and *Aspergillus fumigatus* identified in bronchoalveolar lavage and lung biopsy, is invasive aspergillosis due to prolonged and severe neutropenia, as indicated by the patient's medical history and current laboratory findings, along with the characteristic halo sign on chest CT, suggestive of angioinvasive organisms.

ID: Q112

Question: A 70-year-old man comes to the physician because of a 4-month history of epigastric pain, nausea, and weakness. He has smoked one pack of cigarettes daily for 50 years and drinks one alcoholic beverage daily. He appears emaciated. He is 175 cm (5 ft 9 in) tall and weighs 47 kg (103 lb); BMI is 15 kg/m². He is diagnosed with gastric cancer. Which of the following cytokines is the most likely direct cause of this patient's examination findings?

Expected Answer: IL-6

Generated Answer: : The most likely direct cause of the patient's examination findings, including emaciation and a BMI of 15 kg/m², is the cytokine Tumor Necrosis Factor-alpha (TNF- α). This conclusion is based on the clinical context, which states that cancer cachexia, characterized by weight loss and anorexia, results from a catabolic state related to the interaction between tumor byproducts and the release of cytokines by the host, with TNF- α being one of the pro-inflammatory cytokines secreted by macrophages, leading to weight loss, anorexia, and protein and fat breakdown in cancer patients.

ID: Q113

Question: A 67-year-old woman with advanced bladder cancer comes to the physician for a follow-up examination. She is currently undergoing chemotherapy with an agent that forms cross-links between DNA strands. Serum studies show a creatinine concentration of 2.1 mg/dL and a blood urea nitrogen concentration of 30 mg/dL. Urine dipstick of a clean-catch midstream specimen shows 2+ protein and 1+ glucose. Prior to initiation of chemotherapy, her laboratory values were within the reference range. In addition to hydration, administration of which of the following would most likely have prevented this patient's current condition?

Expected Answer: Amifostine

Generated Answer: Amifostine would most likely have prevented this patient's current condition. As mentioned in Source 1, Kemp et al.'s phase III trial demonstrated that amifostine, given at 910 mg/m² before each chemotherapy cycle, reduced the severity of renal toxicity in patients receiving cisplatin for advanced ovarian cancer. Amifostine is currently marketed for this indication due to its ability to provide statistically significant protection against cisplatin-induced nephrotoxicity, as noted in Source 4.

ID: Q114

Question: A 57-year-old post-menopausal woman comes to the physician because of intermittent, bloody post-coital vaginal discharge for the past month. She does not have pain with intercourse. Eleven years ago, she had LSIL on a routine Pap smear and testing for high-risk HPV strains was positive. Colposcopy showed CIN 1. She has not returned for follow-up Pap smears since then. She is sexually active with her husband only, and they do not use condoms. She has smoked half a pack of cigarettes per day for the past 25 years and does not drink alcohol. On speculum exam, a 1.4 cm, erythematous exophytic mass with ulceration is noted on the posterior wall of the upper third of the vagina. Which of the following is the most probable histopathology of this mass?

Expected Answer: Squamous cell carcinoma

Generated Answer: The most probable histopathology of this mass is squamous cell carcinoma of the vagina. This is supported by the patient's history of abnormal cervical neoplasia, her age, and the presence of an exophytic mass with ulceration on the posterior wall of the upper third of the vagina, which is consistent with squamous cell carcinoma, the most common histologic subtype (85%) of vaginal cancer.

ID: Q115

Question: A 45-year-old man with HIV comes to the physician because of multiple lesions on his chest and lower extremities. The lesions have progressively increased in size and are not painful or pruritic. Current medications include abacavir, dolutegravir, and lamivudine. A photograph of the lesions is shown. His CD4+ T-lymphocyte count is 450/mm³ (normal $\geq$ 500/mm³). A skin biopsy shows multiple spindle-shaped cells and lymphocytic infiltrate. Which of the following is the most appropriate pharmacotherapy?

Expected Answer: Alpha-interferon

Generated Answer: : The most appropriate pharmacotherapy for this patient would be antiretroviral therapy (ART) intensification with a switch to a more potent regimen to achieve better virologic control and immune reconstitution. The current regimen may not be adequately controlling his HIV infection, as evidenced by his CD4+ T-lymphocyte count being below the normal range. Intensification of ART can help improve immune function and potentially reduce the progression and growth of Kaposi's sarcoma (KS) lesions. Additionally, monitoring and management of opportunistic infections, growth factors, nutritional support, and emotional support should also be included in his overall care plan.

ID: Q116

Question: A 61-year-old man with a history of stage IIIa lung adenocarcinoma that has been treated with wedge resection and chemotherapy presents to the primary care clinic. He is largely asymptomatic, but he demonstrates a persistent microcytic anemia despite iron supplementation. Colonoscopy performed 3 years earlier was unremarkable. His past medical history is significant for diabetes mellitus type II, hypertension, acute lymphoblastic leukemia as a child, and hypercholesterolemia. He currently smokes 1 pack of cigarettes per day, drinks a glass of pinot grigio per day, and currently denies any illicit drug use. His vital signs include: temperature, 36.7°C (98.0°F); blood pressure, 126/74 mm Hg; heart rate, 87/min; and respiratory rate, 17/min. On physical examination, his pulses are bounding, complexion is pale, but breath sounds remain clear. Oxygen saturation was initially 91% on room air, with a new oxygen requirement of 2 L by nasal cannula. Which of the following lab values would suggest anemia of chronic disease as the underlying etiology?

Expected Answer: Decreased serum iron and transferrin, increased ferritin, normal serum transferrin receptor

Generated Answer: Anemia of chronic disease in this 61-year-old male patient with stage IIIa lung adenocarcinoma would be suggested by laboratory findings of low serum iron levels, normal or elevated ferritin levels, normal iron stores, and a low serum erythropoietin (EPO) level. This is based on the provided context from Cancer Principles and Practice of Oncology 6e, which describes normochromic, normocytic anemia or anemia of cancer as a common paraneoplastic syndrome characterized by these specific lab results.

ID: Q117

Question: A 62-year-old woman presents to her physician with a painless breast mass on her left breast for the past 4 months. She mentions that she noticed the swelling suddenly one day and thought it would resolve by itself. Instead, it has been slowly increasing in size. On physical examination of the breasts, the physician notes a single non-tender, hard, and fixed nodule over left breast. An ultrasonogram of the breast shows a solid mass, and a fine-needle aspiration biopsy confirms the mass to be lobular carcinoma of the breast. When the patient asks about her prognosis, the physician says that the prognosis can be best determined after both grading and staging of the tumor. Based on the current diagnostic information, the physician says that they can only grade, but not stage, the neoplasm. Which of the following facts about the neoplasm is currently available to the physician?

Expected Answer: The tumor cells exhibit marked nuclear atypia.

Generated Answer: Based on the provided information, the physician can currently determine that the neoplasm is lobular carcinoma of the breast, which tends to be multicentric and commonly bilateral (about 30%). The risk for developing cancer is 20% to 30% in the affected breast and 15% to 20% in the contralateral breast. About 25% to 30% of patients with lobular carcinoma in situ (LCIS) develop ductal carcinoma in situ (DCIS) over 25 to 30 years. The physician can grade the tumor, but not stage it yet, as staging requires axillary lymph node evaluation, which has not been performed.

ID: Q118

Question: A 74-year-old female is brought to the emergency department because of a 2-week history of increasing weakness and chills. She also notes difficulty breathing for the last three days. Eight weeks ago, she underwent left hemicolectomy for adenocarcinoma of the colon.

She subsequently developed a severe urinary tract infection, was treated in the intensive care unit for four days, and was discharged from the hospital three weeks ago. She has type 2 diabetes mellitus, osteoporosis with lumbar pain, hypertension, and atrial fibrillation. She has smoked one pack of cigarettes daily for 50 years. She does not drink alcohol and has never used illicit drugs. Current medications include warfarin, metformin, lisinopril, and aspirin. She appears lethargic and has a large conjunctival hemorrhage in her left eye. Her temperature is 39.3°C (102.7°F), pulse is 112/min, respirations are 25/min, and blood pressure is 126/79 mm Hg. Cardiac auscultation reveals a new holosystolic murmur over the apex. Abdominal examination shows mild, diffuse tenderness throughout the upper quadrants and a well-healed 12-cm paramedian scar. There are multiple tender nodules on the palmar surface of her fingertips. Funduscopic examination shows retinal hemorrhages with pale centers. An ECG shows atrial fibrillation and right bundle branch block. Which of the following is the most likely underlying etiology of this patient's condition?

Expected Answer: Enterococcus faecalis infection

Generated Answer: The most likely underlying etiology of this patient's condition is infective endocarditis, given the presence of a new holosystolic murmur over the apex, multiple tender nodules on the palmar surface of her fingertips (Osler's nodes), and retinal hemorrhages with pale centers (Roth's spots). The patient's recent history of severe urinary tract infection and prolonged hospital stay with intensive care treatment increases her risk for developing septic emboli. Additionally, her atrial fibrillation and history of smoking place her at higher risk for cardiovascular complications. The conjunctival hemorrhage may be related to thrombocytopenia secondary to her malignancy or its therapy.
